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REGULAR PAPER

An Open-source UAV Digital Twin framework: A Case Study on Remote Sensing in the Andean Mountains

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Abstract
The increasing demand for unmanned aerial vehicles (UAVs) in the aerospace industry highlights the need for precise simulation environments, especially in remote regions. This study develops an open-source framework for a customized UAV simulation environment using ROS-Gazebo and Ardupilot. The method includes a realistic reconstruction of the environment based on free satellite information, the ROS-Gazebo scheme for modeling and testing the UAV platform, and Ardupilot for mission design and deployment. The methodology aims to reduce costs in surveillance missions and conduct more efficient operations in high-risk and challenging areas. To evaluate the accuracy of UAV positioning for simulating real missions, an experimental validation of the digital twin was carried out using real flight data records from two self-built UAVs equipped with the open-source Ardupilot autopilot. One of the aircraft, a fixed-wing UAV, was tested near the Antisana volcano (4500 meters above sea level), where wind gusts reached speeds of 9 to 12 m/s. These tests revealed maximum errors of 9% in the Z-axis trajectory (altitude). The other aircraft, a quadcopter, was evaluated in the Parque Carolina in Quito (2800 meters above sea level), with wind gusts between 5 and 7 m/s, showing an error of 8% in the Z-axis trajectory. The presented results demonstrate the suitability of the proposed method for emulating complex missions using digital twin models. In this regard, the main contribution of this work lies in its potential for the precise prediction of flight missions in the Z-axis, a crucial variable for avoiding collisions in mountainous regions.

Keywords Ardupilot · Digital twin · ROS · Gazebo · Fixed-Wing UAV · Opensource framework

1 Introduction

The increasing demand for Unmanned Aerial Vehicles (UAVs) for monitoring extensive and high-interest regions has driven the development of more precise and secure platforms. A notable example of these areas is the high-Andean wetlands in Ecuador, which have deteriorated due to human intervention and climate change [1, 2]. The degradation of these wetlands has a negative impact on the local ecosystem and the communities dependent on water sources originating from these areas [3]. Therefore, it is crucial to develop non-invasive monitoring and analysis tools to track the degenerative mechanisms affecting these ecosystems.

Studies by Valencia et al. [4–6] suggest that fixed-wing UAVs are a viable alternative for monitoring large areas of Andean wetlands, thanks to their high-resolution imagery and wide coverage. These UAVs excel in their ability to generate lift due to their wingspan, providing superior autonomy ideal for extensive monitoring missions [7–9]. Additionally, their aerodynamics enable them to withstand strong wind gusts common in the high Andes [4, 5, 10, 11]. However, their operation is challenging and manufacturing costs are high, which hinders their use in flight tests for data collection in research activities [12–14]. As a result, quadcopters

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are often preferred, limiting the development of digital systems for other types of aircraft.

In this context, the development of Digital Twins (DTs) offers a way to test different propulsion architectures and aerodynamic structures, evaluate new flight control systems, or assess new payloads and sensors [15]. DTs also help reduce risks to personnel, prevent potential damage to UAV structures, and lower logistical and monitoring equipment costs. These advantages are outlined in [16], which proposes a Software-in-the-Loop (SITL) simulation using an open-source framework for a quadcopter. The focus is on basic stabilization and control within a simulated environment, without including detailed aerodynamic modeling or lift and drag coefficients obtained through CFD. In a more physically integrated approach, [17] develops a digital twin platform for multirotor UAVs. This platform integrates real-time sensor data to improve interaction between the physical and virtual environments. Although it performs well in urban scenarios, the study does not consider fixed-wing aircraft or wind conditions. Complementarily, [18] suggests using Gazebo, ROS, and QGroundControl to emulate UAVs while tuning PID controllers. This type of simulation has been adopted in studies that test multiple UAV platforms simultaneously [19–21]. However, these works do not include full aerodynamic characterization or simulate missions with physical realism.

For fixed-wing UAVs, Zhang et al. [22] present FlightTwin, a generalized digital twin framework directly integrated with the flight system. Their approach focuses on system modularity and replicating the UAV's physical behavior under nominal conditions. However, the study does not address non-ideal flight scenarios such as gusts or severe external disturbances. Aláez et al. [23] consider these conditions in their proposal. They present a digital twin for VTOL UAVs focused on takeoff, hovering, and landing under different wind conditions. Although their model is robust, it is limited to stationary flight phases and does not include full mission trajectory analysis.

SITL models have also been used to test advanced control strategies, such as adaptive neuro-fuzzy attitude control, nonlinear flight controllers, and improved trajectory tracking approaches in virtual environments [24–28]. Other studies have explored the impact of external disturbances on the effectiveness of controllers, simulating more realistic flight conditions [29, 30]. However, most previous research has evaluated DTs under ideal conditions, without considering external disturbances such as wind, a critical factor in simulating the real behavior of UAVs in mountainous environments [31]. Fixed-wing UAVs, in particular, are significantly more affected by wind due to their aerodynamic configuration, requiring more sophisticated control strategies to maintain stability in turbulent conditions. While some studies have addressed trajectory optimization under wind disturbances, no digital twin framework has yet been developed to explicitly model these effects on a fixed-wing UAV.

This work addresses these gaps by developing a digital twin for UAV simulation in high-altitude environments, characterized by challenging atmospheric conditions such as strong winds and turbulence. To achieve this, an open-source simulation framework is integrated to model wind effects on the aircraft. This framework is based on real flight data obtained during missions conducted in mountainous regions. Thus, this DT can be used to more accurately assess UAV performance in strong wind environments before conducting tests with a real prototype. In this way, the risk of flight accidents, which could negatively impact the surrounding environment, is reduced.

It is important to note that, in this context, the term DT refers to the virtual replica or model of a physical entity [32]. This concept encompasses three key ideas: observing physical products under real-world conditions, representing physical twins in virtual space, and establishing correlations between simulated models and reality, as described in [33]. Its application in UAV simulation allows for a realistic evaluation of control strategies, different aircraft models, and the prediction of relevant events [34, 35].

The remainder of this document is organized as follows: Chapter 2 describes the methodology used for developing the digital twin. This chapter includes the implementation of the aerodynamic model, the monitoring area, and flight simulations. Chapter 3 presents the obtained results and their comparison with experimental data for both a fixed-wing aircraft and a quadcopter. In addition, the results obtained for the two types of aircraft are discussed, and the limitations of the implemented framework are determined.

2 Methodology

The customized simulation environment designed to evaluate the fixed-wing DT replicates intricate geographic terrains and wind perturbations. This methodology incorporates Ardupilot's Software-in-the-Loop (SITL) framework to emulate communication, control, and mission planning through various open-source software tools. A comprehensive overview of the framework is provided in Fig. 1. Within this methodology, Ardupilot emulates the open-source flight controller Pixhawk, while QGroundContol (QGC) [16] is used for mission planning, data logging, sensor testing, and related tasks. Additionally, the middleware Robot Operating System (ROS), in conjunction with Gazebo, manages physics, model aerodynamics, and actuator control. ROS also oversees communication protocols, connecting the Ardupilot controller and QGC through MAVROS nodes and the MAVlink protocol.

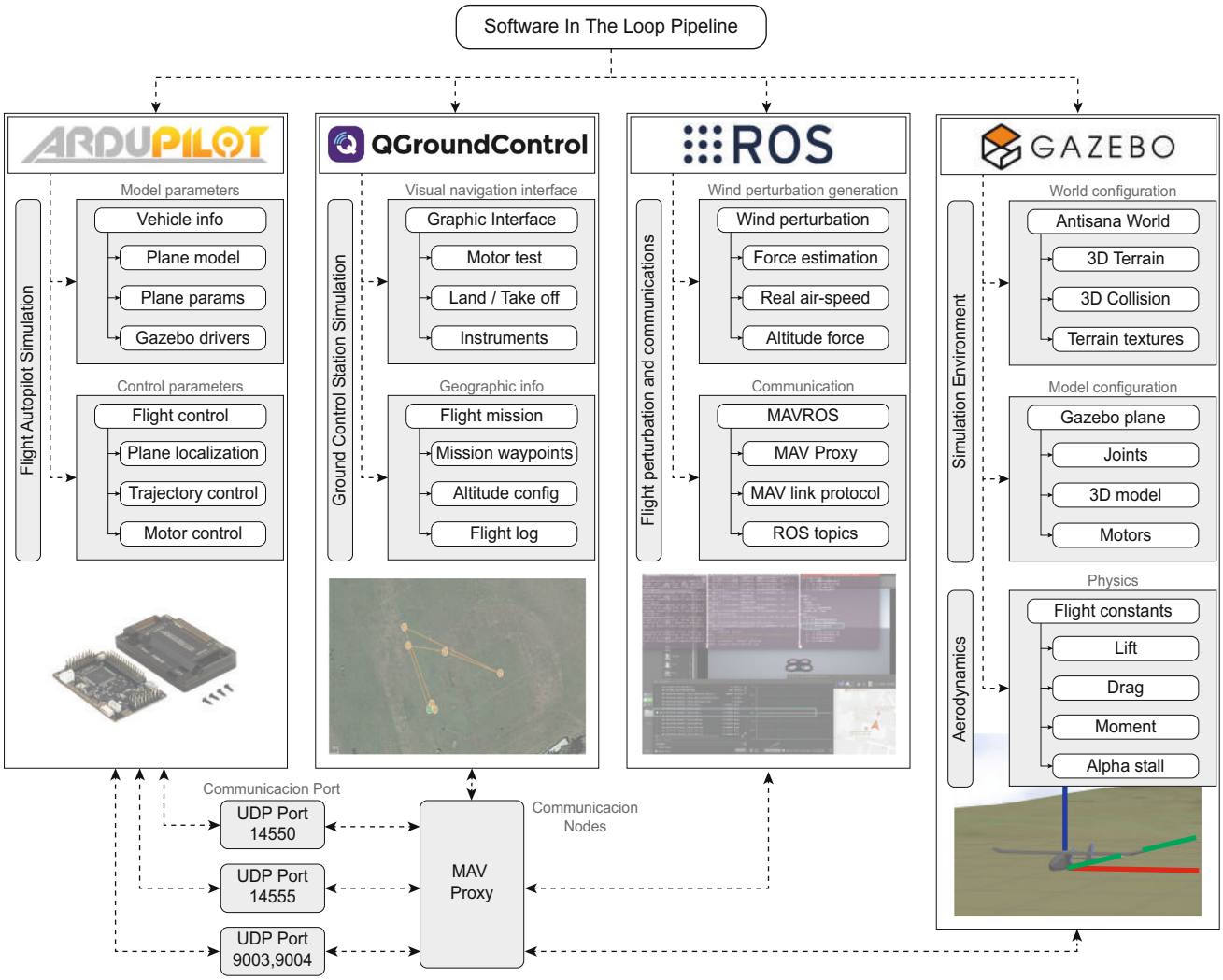

Fig. 1 Simulation scheme that combines Ardupilot, ROS, Gazebo and QGroundControl, and the UAV Digital Twin

These tools enable the development of a digital testing scenario for a digital fixed-wing aircraft. To enhance the realism of the virtual scenario, an approach is proposed to extract three-dimensional information from the test terrain, specifically the surroundings of the testing site (Antisana volcano), and integrate it into the Gazebo environment. Moreover, this approach involves a ROS node that estimates a perturbation force caused by strong wind gusts, which is actively applied to the DT. This node utilizes altitude data from actual flights to generate a single-axis perturbation force, replicating wind conditions at the time of the real data collection.

The proposed approach was validated against real flight data obtained during a surveillance mission using a fixed-wing aircraft. Additionally, as mentioned in the introduction, to test the versatility of the presented method, the work also showcases a quadrotor tested in open spaces within an urban scenario. The experiments with the quadrotor are derived from a conference paper titled "Open-Source Software-in-the-Loop Strategies for Realistic UAV Monitoring Applications" [36]. For the benefit of the community, the detailed installation guide for the open-source simulation tools used in this work can be found at https://gi-ata.gitbook.io/ros-arducopter. Furthermore, data logs from the quadrotor and fixed-wing experiments can be accessed at https://github.com/estebanvt/OSSITLQUAD. In the following subsections, each step of the SITL open-source framework is described.

2.1 3D Height Map Modelling

The starting point for obtaining an accurate simulation environment involves acquiring spatial information for monitoring purposes. In the context of this research, which focuses on a complex topological environment such as the Ecuadorian highlands, significant challenges arise. The uncertainty in the z-axis, generated by the GPS measurement system as

indicated in [37], and the lack of access to freely available satellite images for the equatorial region [37], cause errors in the order of centimeters (cm) for the Digital Terrain Model (DTM).

The flight area selected for this research is located in the Ecuadorian Andean region, with geographic coordinates Universal Transverse Mercator (UTM) 0°30'05.4" S 78°13'08.6" W. Specifically, the flight zone is within the Antisana National Park, whose altitude ranges from 1,200 to 5,758 meters above sea level. The area of interest for the flight is located around the Antisana volcano, which is at an altitude of 4,500 meters above sea level (masl). The difficult access in this region increases the risks and operational costs for non-governmental organizations (NGOs) working in wetland conservation. Low-cost UAVs used for wetland conservation are often not well-equipped for such missions. Therefore, the ATA research group at the National Polytechnic School has developed a fixed-wing UAV designed to operate in challenging conditions, including low temperatures, strong wind gusts, and low air density [5, 38].

In addition to the rugged terrain morphology, this area experiences significant rainfall and snowfall from May to October. Precipitation plays a crucial role in shaping the landscape and the ecosystem of the volcano, influencing water sources and contributing to unique environmental conditions in the region [39]. The choice of the Antisana sector is due to ongoing research projects where the health of the wetlands is periodically monitored [40].

The methodology for retrieving the three-dimensional information for the simulation environment is summarized in Fig. 2, and detailed in the following subsections.

2.1.1 Landscape Reconstruction

As mentioned in the previous section, one of the main objectives of this work is to obtain highly accurate terrain topography. To achieve this, satellite images are used to generate the three-dimensional scene that the DTM will interact with. Although these images have errors of +/-10 meters [38], their free access favors the affordability of the method in monitoring missions. There are several open-source satellite image datasets available, such as Copernicus Open Access Hub (ESA), EarthData search (NASA), and USGS Earth Explorer. These datasets offer a variety of images, including multispectral, panchromatic, and hyperspectral, which are essential for accurate topographic reconstruction. However, using such datasets requires a careful review of their spatial and temporal resolution, as well as potential issues with missing data or cloud cover.

To address the challenges of missing data, preprocessing steps were applied to clean the datasets. Images with missing or incomplete data were removed to ensure the quality and accuracy of the final topographic model. Additionally, a mosaic was created by combining multiple images to cover the area of interest uniformly. Furthermore, images with minimal cloud cover were selected, as cloud cover affects the accuracy of hyperspectral sensors and, consequently, the precision of the 3D model.

The digital elevation model for Ecuador that best suits our needs was obtained from the Shuttle Radar Topography Mission (SRTM). This is a NASA project focused on creating three-dimensional models of specific areas [38, 41]. The next step is to use the satellite height map information and transform it into an STL model using the BlenderGIS plu-

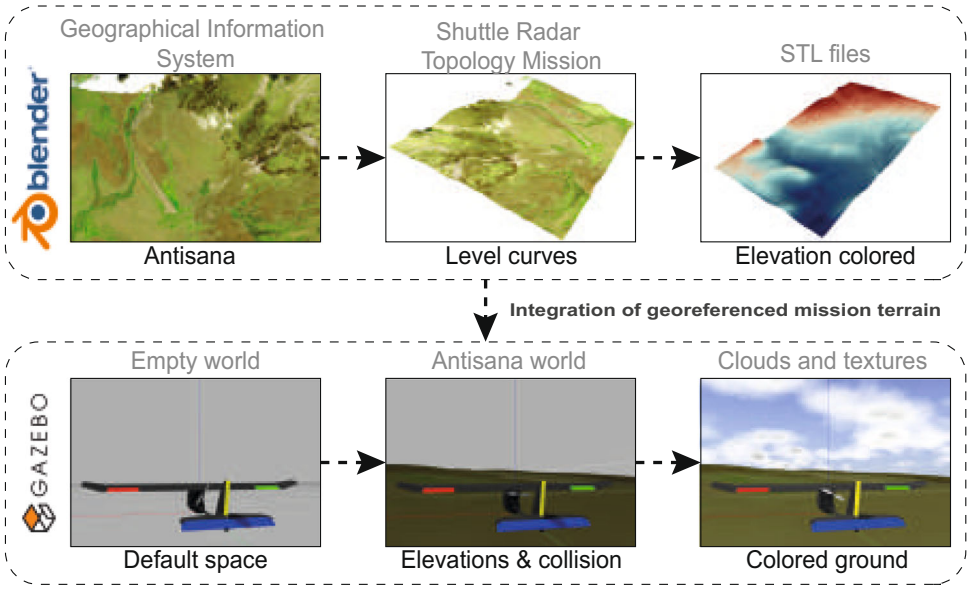

Fig. 2 Methodology for integration of geo-referenced terrain with elevations into Gazebo

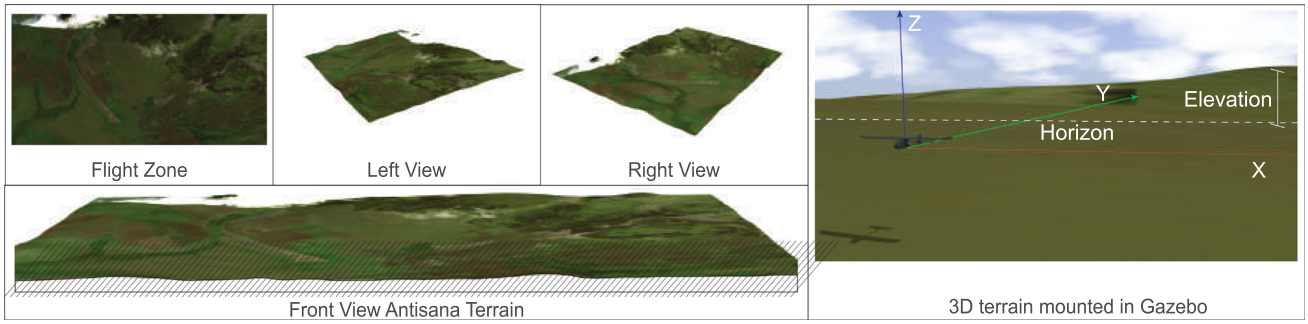

Fig. 3 Views of the Antisana terrain mesh and fixed-wing unmanned aircraft assembled in Gazebo

gin. BlenderGIS enhances Blender's capabilities by allowing the import of geographic data and 3D visualization of Geographic Information System (GIS) data. It also provides tools for spatial analysis and options to export models and data to formats used in GIS, integrating GIS-related tasks into Blender workflows [42].

One of the most significant reasons for opting for BlenderGIS arose from previous collaboration with QGIS, an open-source Geographic Information System licensed under GNU. This phase of work with QGIS was pivotal in the final selection of the plugin. However, two challenges arose in QGIS: the dimensions of the provided STL file, which required scaling adjustments, and the lack of texture for the flight zone.

Spatial information about terrain, such as elevation, soil morphology, and texture, is crucial for realistically emulating the environment where the fixed-wing digital twin will perform controlled missions. After selecting the area of interest via BlenderGIS, elevation data such as contour lines can be retrieved [43]. The result of this operation can be seen in Fig. 3, where a flight zone in Antisana National Park is recovered. Contour lines serve as collision guides when exporting this environment to Gazebo. It is important to note that cloud cover can affect terrain visualization. Therefore, to improve the texture of the recovered 3D model, map textures exported from Google Earth satellite images can be used and integrated into BlenderGIS.

2.1.2 Environment Emulation in Gazebo

The implementation of the DT model in Gazebo initially occurs in a default empty environment. After retrieving the 3D terrain information, as shown in Fig. 2, this data is integrated into Gazebo. This allows the terrain elevation to be perceptible to the operator, and the unmanned aircraft can interact with elevated areas thanks to collisions calculated with contour lines. In the Gazebo scenario, clouds are included exclusively to enhance maneuverability and the ability to distinguish high elevations during manual operations, as illustrated in Fig. 3. The Gazebo .world file is configured for this purpose, allowing the representation of clouds to be disabled if computational resources are limited.

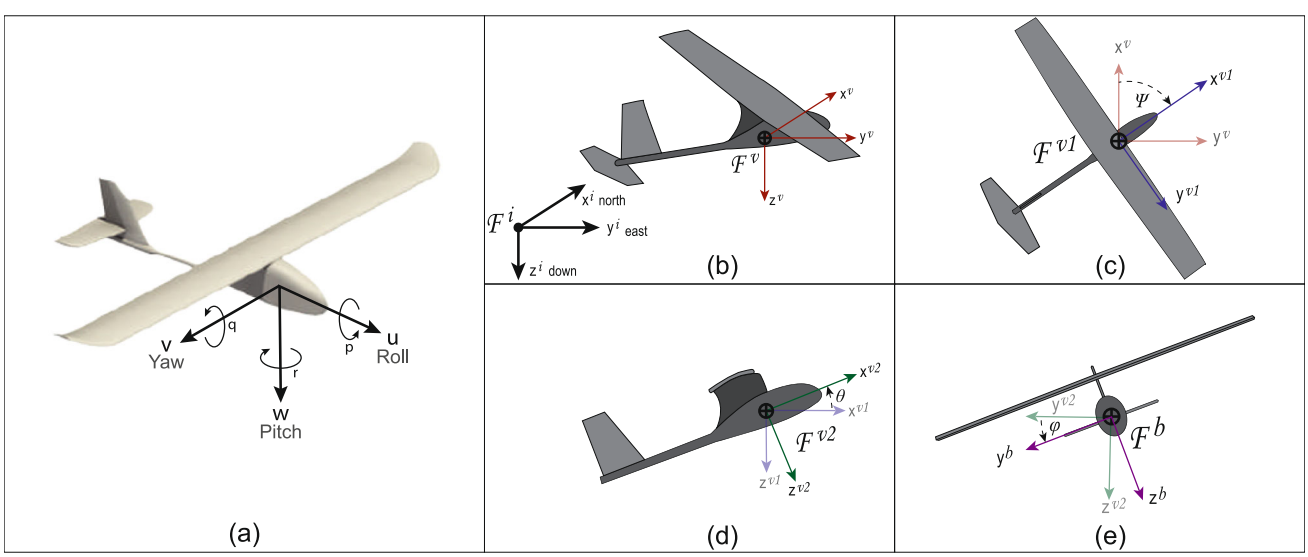

Fig. 4 Fixed wing aircraft axes of motion and frames: (a) Velocities around each axis, (b) frame *v* (c) frame *v1*, (d) frame *v2*, (e) frame *b*

2.2 UAV Characterization

The following paragraphs present a brief summary of fixed-wing plane modeling, which is fully detailed in [44].The equations presented below allow us to understand the external forces and moments that should be characterized and properly configured in the simulation environment.

The model state terms are defined in Table 1, and serve as means to explain the kinematic and dynamic model of the six-degree-of-freedom fixed-wing aircraft, which is summarized in the following equations.

The translational velocity in frame v can be computed from velocities stated in b frame yielding, on the other hand, the body-frame angular rates can be expressed in terms of the derivatives of the Euler angles given the proper rotational transformations as follows,

$$\begin{pmatrix} \dot{p}_n \\ \dot{p}_e \\ \dot{p}_d \end{pmatrix} = \begin{pmatrix} \cos\theta\cos\psi & \sin\phi\sin\theta\cos\psi - \cos\phi\sin\psi & \cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi \\ \cos\theta\sin\psi & \sin\phi\sin\theta\sin\psi + \cos\phi\cos\psi & \cos\phi\sin\theta\sin\psi - \sin\phi\sin\psi \\ -\sin\theta & \sin\phi\cos\theta & \cos\phi\cos\theta \end{pmatrix} \begin{pmatrix} u \\ v \\ w \end{pmatrix}, \tag{1}$$

2.2.1 Mathematical Model

This section summarizes the mathematical model of the unmanned aircraft, where its translational and rotational dynamics are driven by external forces and moments due to gravity and aerodynamics. The model follows the methodology proposed in [44] taking into account the axes definition (roll, pitch, and yaw) stated in Fig. 4a.

It is denoted $\mathcal{F}^i$ as the inertial frame (the static world origin reference), $\mathcal{F}^v$ is the vehicle frame parallel to the inertial frame placed in the aircraft's center of mass, see Fig. 4b. The frame $\mathcal{F}^{v1}$ in Fig. 4c is rotated ψ degrees around z^v pointing out the nose of the aircraft. The frame $\mathcal{F}^{v2}$ is obtained by rotating $\mathcal{F}^{v1}$ by an angle θ around the y^{v1} axis, which corresponds to the pitch axis of the vehicle. This transformation allows the vehicle's orientation to be adjusted with respect to the initial body frame $\mathcal{F}^{v1}$, as shown in Fig. 4c. The rotation of the coordinate system follows the Euler Angles, which describe the orientation of a rigid body in space with respect to a fixed coordinate system. Finally, frame $\mathcal{F}^b$ -or body frame- is another rotation of $\mathcal{F}^{v2}$ by ϕ around x^{v2} as shown in Fig. 4e.

$$\begin{pmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{pmatrix} = \begin{pmatrix} 1 & \sin\phi\tan\theta & \cos\phi\tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi/\cos\theta & \cos\phi/\cos\theta \end{pmatrix} \begin{pmatrix} p \\ q \\ r \end{pmatrix}. \tag{2}$$

In the case of an aircraft in maneuver, the application of the second law of Newton can be easily done by expressing forces and velocities in the body frame that yields,

$$\begin{pmatrix} \dot{u} \\ \dot{v} \\ \dot{w} \end{pmatrix} = \begin{pmatrix} rv - qw \\ pw - ru \\ qu - pv \end{pmatrix} + \frac{1}{m} \begin{pmatrix} f_x \\ f_y \\ f_z \end{pmatrix}, \tag{3}$$

where, the external forces (f_x, f_y, f_z) represent the aerodynamic and gravitational forces acting on the aircraft, and are expressed in the body frame $\mathcal{F}^b$. These forces include lift, drag, and thrust, as well as any perturbations due to gravity. The term m denotes the mass of the aircraft, and the forces are normalized by this mass to obtain the accelerations in the body frame. This equation describes the translational motion of the aircraft, accounting for both its velocity in the body frame and the external forces acting on it. Finally, Eq. 4

Table 1 State components for the fixed-wing unmanned aircraft [44]

Variable	Description	Units
p_n	Inertial north position along roll axis in $\mathcal{F}^i$	[m]
p_e	Inertial east position along pitch axis in in $\mathcal{F}^i$	[m]
p_d	Inertial down position along yaw axis in $\mathcal{F}^i$	[m]
u	Velocity measured along roll axis in $\mathcal{F}^b$	[m/s]
v	Velocity measured along pitch axis in $\mathcal{F}^b$	[m/s]
w	Velocity measured along yaw axis in $\mathcal{F}^b$	[m/s]
ϕ	Roll angle defined with respect to F^{v2}	[rad]
θ	Pitch angle defined with respect to F^{v1}	[rad]
ψ	Heading (yaw) angle defined with respect to F^v	[rad]
p	Roll rate measured along roll axis	[rad/s]
q	Pitch rate measured along pitch axis	[rad/s]
r	Yaw rate measured along yaw axis	[rad/s]

expresses the relationship between the externally applied moments and the rotational dynamics of the aircraft, incorporating the moment of inertia tensor $\mathbf{J}$ in the body frame:

$$\begin{pmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{pmatrix} = \begin{pmatrix} \Gamma_1 pq - \Gamma_2 qr \\ \Gamma_5 pr - \Gamma_6 (p^2 - r^2) \\ \Gamma_7 pq - \Gamma_1 qr \end{pmatrix} + \begin{pmatrix} \Gamma_3 l + \Gamma_4 n \\ \frac{1}{J_y} m \\ \Gamma_4 l + \Gamma_8 n \end{pmatrix} \text{ with } \mathbf{J} \triangleq \begin{pmatrix} J_x & -J_{xy} & -J_{xz} \\ -J_{xy} & J_y & -J_{yz} \\ -J_{xz} & -J_{yz} & J_z \end{pmatrix}, \tag{4}$$

with the following gamma terms:

$$[l]\Gamma_1 = \frac{J_{xz}(J_x - J_y + J_z)}{\Gamma}, \quad \Gamma_2 = \frac{J_z(J_z - J_y) + J_{xz}^2}{\Gamma}, \quad \Gamma_3 = \frac{J_z}{\Gamma}$$
$$\Gamma_4 = \frac{J_{xz}}{\Gamma}, \quad \Gamma_5 = \frac{J_z - J_x}{J_y}, \quad \Gamma_6 = \frac{J_{xz}}{J_y}.$$
$$\Gamma_7 = \frac{(J_x - J_y)J_x + J_{xz}^2}{\Gamma}, \quad \Gamma_8 = \frac{J_z}{\Gamma}, \quad \Gamma = J_x J_z - J_{xz}^2$$

The Gamma terms represent dimensionless coefficients that are derived from the inertia tensor $\mathbf{J}$. These terms are used to express the rotational dynamics of the aircraft, where J_x, J_y, J_z are the moments of inertia about the principal axes, and J_{xy}, J_{xz}, J_{yz} are the products of inertia. The term Γ is the determinant of the inertia matrix, and it plays a critical role in defining the rotational resistance of the aircraft.

2.2.2 Aerodynamic Characterization of the UAV

The comprehensive study of aerodynamic forces on small fixed-wing UAVs, presented in Narváez et al. [45] serves as a foundation for understanding UAV behavior during monitoring missions in the high Andean wetlands. The geometric characteristics of the simulated fixed-wing aircraft are presented in Table 2. These characteristics were derived from an internally built aircraft model, which was used in subsequent experiments for validation purposes.

To emulate realistic aerodynamic behaviors, Gazebo utilizes a specialized lift and drag plugin. This component is crucial as it introduces aerodynamically relevant behaviors into simulated objects, including lift and drag coefficients [46].

Table 2 Fixed-wing geometric features [45]

Dimension	Value	Units
Wingspan	2.20	[m]
Chord	0.235	[m]
UAV reference area	0.5125	[m^2]
Wing reference area	0.3997	[m^2]
Weight	2.5	[kg]
Cruise speed	12.0	[m/s]
Flight autonomy	85.0	[min]
Max Payload	0.5	[kg]

The accuracy of this plugin's performance depends on the quality of various coefficients related to UAV geometry and lift and drag forces. This allows for proper aerodynamic characterization and a deeper understanding of how the UAV responds to variable conditions when the Angle of Attack (AoA) changes. AoA analysis for this fixed-wing aircraft was conducted in the range of -10 to 30 degrees. Within this range, lift and drag coefficients were discretely calculated using computational fluid dynamics (CFD) simulations with the OpenFOAM platform [47], as depicted in Fig. 5. The calculation of UAV geometric coefficients follows the approach described in [46], based on slope estimation for lift and drag, as shown in Fig. 5. The resulting coefficients are used in all simulations within Gazebo and are summarized in Table 3.

The fidelity achieved in the simulations depends on a meticulous numerical approach to the complex UAV geometry, which involves CFD simulations using OpenFOAM. This numerical approach requires the construction of a structured C-type mesh configuration using the 'blockMesh' utility within OpenFOAM. This mesh configuration provides the necessary flexibility to adjust wind direction parameters for different AoAs without the need to generate new meshes for each case. The mesh creation process follows a systematic approach using BlockMesh and refinement with SnappyHexMesh, ensuring mesh integrity through continuous monitoring of quality and orthogonality, as described in [47]. The simulations were conducted under turbulent

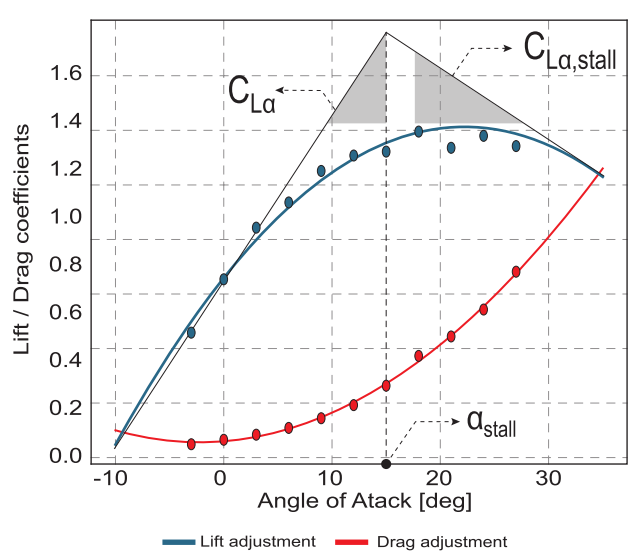

Fig. 5 Forces Coefficients vs AoA

Table 3 Aerodynamic coefficients for the fixed-wing plane obtained by the simulation in OpenFOAM and the Gazebo LiftDrag plugin [46]

Constant name		Value	Units
Zero-lift Angle of Attack	α_0	0.05236	[rad]
Ratio coefficient of Lift and alpha slope before stall	$C_{L\alpha}$	3.50141	[-]
Ratio coefficient of Drag and alpha slope before stall	$C_{D\alpha}$	0.63662	[-]
Ratio coefficient of Moment and alpha slope before stall	$C_{M\alpha}$	−0.2040	[-]
Angle of stall	α_{stall}	0.26180	[rad]
Ratio coefficient of Lift and alpha slope after stall	$C_{L\alpha,\text{stall}}$	−1.1459	[-]
Ratio coefficient of Drag and alpha slope after stall	$C_{D\alpha,\text{stall}}$	2.29183	[-]
Ratio coefficient of Moment and alpha slope before stall	$C_{M\alpha,\text{stall}}$	−0.1313	[-]

flow conditions using the k-ω SST turbulence model, which is widely employed in external aerodynamic applications. Additionally, standard atmospheric conditions at sea level were assumed, in line with the approach followed in [45, 48], where a similar case study is presented for a fixed-wing UAV operating in high Andean wetland monitoring.

After creating the mesh, several CFD simulations are conducted using different solvers such as `simpleFoam` or `pimpleFoam`, allowing for a thorough exploration of aerodynamic flows. The post-processing of the simulations involves fluid dynamics data processing to calculate the force coefficients listed in Table 3.

2.3 Digital Twin Simulation Scheme

The simulation scheme for the UAV's DT requires open-source tools such as Ardupilot, ROS, and QGC. Once Ardupilot is installed on the open-source Ubuntu 20.04 system, the desired aircraft for the simulation must be properly configured by specifying all aerodynamic coefficients. Finally, the aircraft needs to be integrated into the Gazebo environment. This involves creating an *.sdf* file that describes the physical characteristics, components, articulation points facilitating flight, and the inclusion of LiftDrag and communication plugins. The communication protocol between the simulated aircraft and the flight controller, simulated by certain programs within Ardupilot, is MAVLink [49]. MAVLink allows the visualization of the aircraft in Gazebo to connect with the QGC ground station.

Executing the 3D simulation of the aircraft requires ROS to act as a bridge between Gazebo and Ardupilot. The *.launch* files are used to automate the loading of ROS nodes and the simulation setup. In this case, a specific *.launch* file (antisana.launch) initializes a Gazebo world with a predefined flight location near the Antisana volcano. This region presents a challenging environment with variable wind speeds (4 to 14 m/s) at an altitude of approximately 4100 masl. Executing the "Roslaunch" command along with this file opens the Gazebo window with the UAV's physical, aerodynamic, and visual characteristics, ensuring that the simulation accurately reflects its real behavior.

Ardupilot supports the simulation of various types of vehicles, including helicopters, planes, rovers, and submarines. The console for input and output opens at the specified coordinates, in this case, the coordinates of Antisana. When running, the aircraft, using the MAVLink protocol, connects to the console that simulates the flight controller. Opening the QGC ground station with the same MAVLink communication protocol visualizes the connection between the aircraft and the ground station.

With the assistance of QGC, a flight plan or mission is generated, and the aircraft's performance is observed within the Gazebo environment. It is important to note that any external force or disturbance applied in Gazebo is tracked by QGC. Communication is bidirectional, although not all parameters are configurable. Finally, to simulate a flight similar to those performed around Antisana, a *.launch* file is executed, activating a ROS node responsible for introducing a real-time disturbance into the simulation. This node runs a Python script that configures the previously described wind conditions, ensuring that the simulation accurately reflects the real environmental conditions.

2.3.1 Wind Perturbation Modeling

The virtual environment for this approach runs in Gazebo with all necessary features, such as mission planning, trajectory control modules, and 3D scenarios. However, along the predefined path, there are no disturbances, and the trajectory remains unaltered. To have a fully functional digital twin, it is imperative to introduce disturbances that emulate those present in real tests. The most representative external disturbance is wind behavior, which significantly impacts the aircraft's performance during the predefined mission [50].

The disturbance caused by the wind during a mission was modeled as a force. This perturbation force is estimated fol-

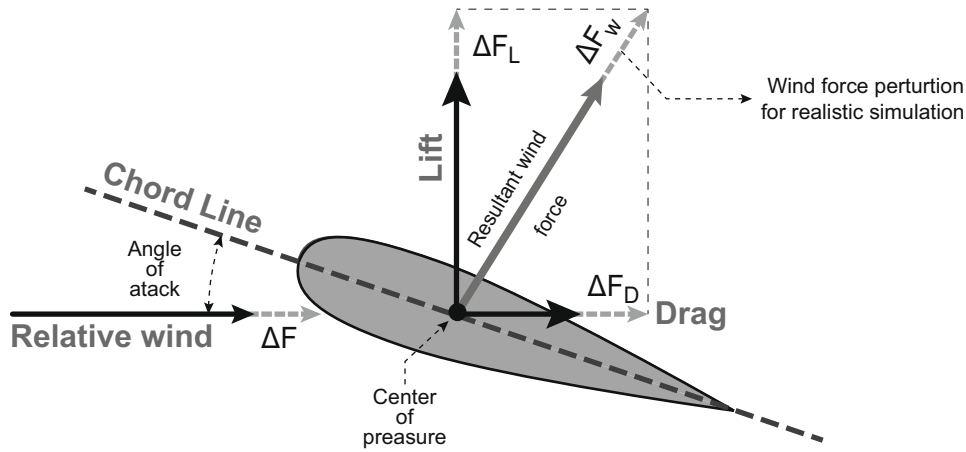

Fig. 6 Force vectors on an airfoil

lowing the lift and drag forces model [51] described in Fig. 6, resulting in:

$$\Delta F_D \approx 0.5 \cdot \rho \cdot A \cdot C_D \cdot V^2 \tag{5}$$

$$\Delta F_L \approx 0.5 \cdot \rho \cdot A \cdot C_L \cdot V^2 \tag{6}$$

$$\Delta F_W = ||F_D + F_L|| \tag{7}$$

where ρ is the air density set as a constant of about 0.83 (kg/m^3), $A(m^2)$ the area of the object exposed to the wind, $V(m/s)$ is the wind speed, C_D and C_L are the dimensionless drag and lift coefficients as stated in Table 3. Recall that the air density is an average value that takes into account the altitude of the Antisana mission terrain. The wind speed V is approximated based on the difference in altitude between the reference in a real flight test, and an adjustment factor tuned heuristically.

As shown in Fig. 6, the resultant wind perturbation force ΔF_W is applied as a force along the Z axis at the center of mass of the UAV. The computed perturbation forces can be used as simulation disruptions for other missions, assuming the same climatic conditions under which the real data were acquired.

In real flight scenarios, flight logs serve as comprehensive data repositories on UAV performance throughout an entire mission. By analyzing these logs, fluctuations in estimated wind speed are tabulated using a simple yet effective approach. This allows for the introduction of a single-axis perturbation force. Real data indicated that most trajectory deviations are concentrated along the Z-axis, while those in the X-Y plane are compensated for by flight control.

Therefore, the effort is focused on estimating a perturbation force along a single axis. The calculated perturbations are applied to the aircraft model using a Python script that discretely publishes these values after takeoff and before landing. It is worth noting that no perturbations were added during the takeoff and landing stages.

The estimation approach described in this section is preliminary work. In future research, the goal is to generate a three-dimensional perturbation from wind fluctuation, similar to highly variable gusts experienced in Andean wetlands.

3 Results and Discussion

The experimental validation of the proposed approach was carried out in the wetlands of Antisana volcano in Ecuador. As mentioned earlier, one of the main objectives of this work was to validate the simulation with real data in open environments with challenging operational conditions. For this task, a custom-built fixed-wing UAV equipped with a Pixhawk flight controller was used. Three flight missions were considered for evaluation, labeled 158, 178, and 185. These

Table 4 Mission features

UAV type	Mission label	Date	Distance of mission	Height above ground level
Fixed-wing	158	02-10-2023	11739 [m]	80[m]
Fixed-wing	178	11-12-2023	11003 [m]	120[m]
Fixed-wing	185	04-02-2023	5846[m]	100[m]
Quadrotor	Carolina-40	29-01-2024	326[m]	40[m]
Quadrotor	Carolina-20	29-01-2024	326[m]	20[m]
Quadrotor	EPN-30	22-01-2024	226[m]	30[m]
Quadrotor	EPN-20	22-01-2024	226[m]	20[m]

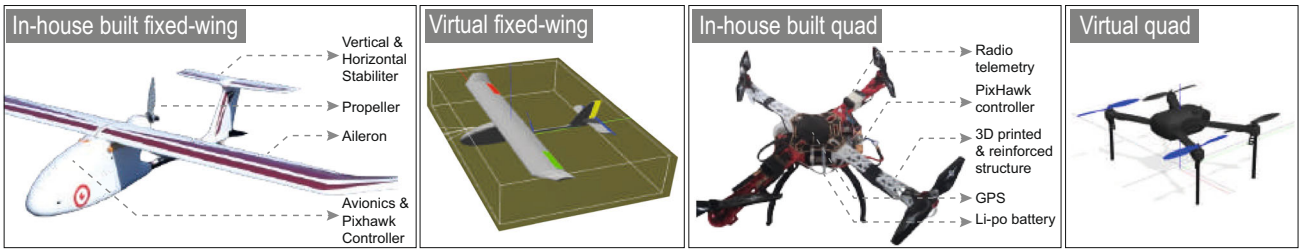

Fig. 7 UAV models used in the validation process along their respective digital twin

missions were conducted on different dates with varying configurations and altitudes to account for different wind and weather conditions, as detailed in Table 4.

Furthermore, to assess the versatility of the developed open-source framework and the estimation of perturbing forces, the approach was implemented on an internally built quadcopter. The UAVs used in these experiments are shown in Fig. 7, along with their respective digital models.

The quadcopter missions were conducted in two open spaces in the city of Quito, at different altitudes known as Carolina and EPN, as described in Table 4. Due to the autonomy and stability limitations of the quadcopter, which affect its performance in strong wind conditions, flight tests were conducted in the mentioned urban environments. This allowed for evaluating the methodology under wind conditions that are more manageable for the quadcopter.

It is important to remember that the experiments with the quadcopter complement the case study in Antisana and are part of the work described in [36]. The following subsections provide details on the experiments with both types of UAVs.

3.1 Fixed-wing Plane Experiments

Flight missions 158 and 178 were conducted during months characterized by stronger wind gusts and challenging weather conditions, according to the INHAMI meteorological information system in Ecuador [52]. Real flights covered an average path length of approximately 11 km, with preconfigured mission shapes illustrated in Fig. 8.

Once the 3D terrain model was established, simulations with the digital twin of the fixed-wing aircraft were conducted. In the absence of external disturbances, the ideal simulation environment resulted in flat and stable trajectories. These trajectories exhibited minor deviations during turns, likely attributed to the Ardupilot controller settings. The ideal simulation results for both experiments are shown in blue in Fig. 8a and b; both demonstrate similar performance despite the different altitudes of each test.

By using altitude variations derived from real flight data, a disturbance force along a single axis that closely mimics wind effects during the execution of both experiments was estimated. The results manifested in highly irregular trajectories, illustrated in red in Fig. 8a and b. For a clearer perspective on these simulations, readers are encour-

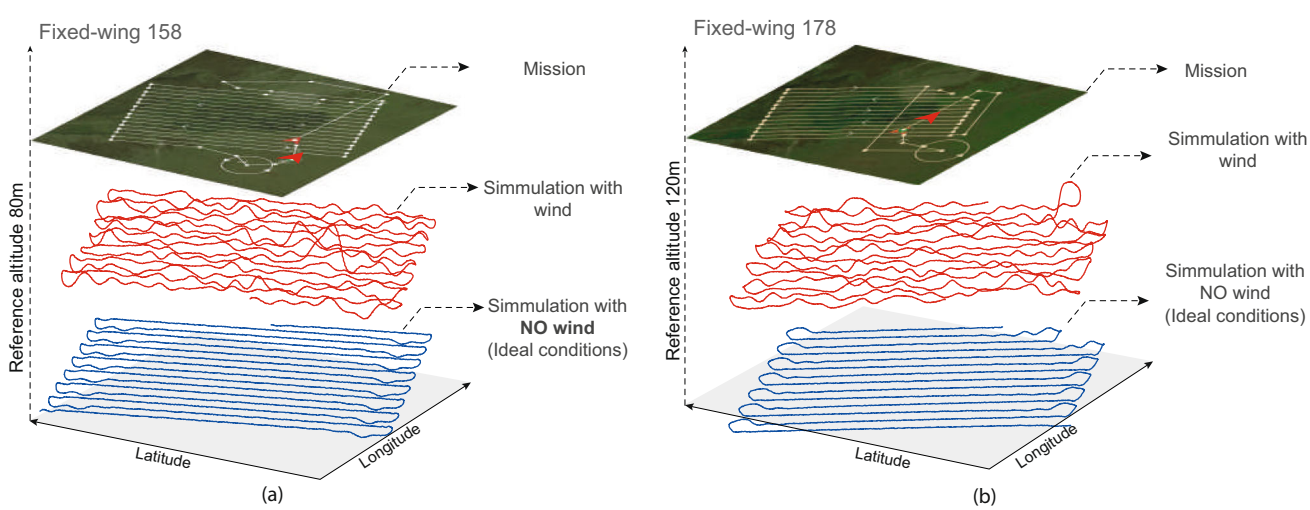

Fig. 8 Flight tests using a fix with and without wind perturbation: (a) experiment 158, (b) experiment 178

aged to view the accompanying video at https://youtu.be/-BWEreZNBy8?si=UUQQ7rVRCC_j3qKr.

The validation process involves heuristic adjustment of constants to produce an estimation of disturbance that generates altitude values as close as possible to those from real flight tests. Fine-tuning the disturbance force allowed precise altitude estimations with the digital twin, as shown in the comparisons between real and simulated data in Fig. 8a and b.

Applying an external force entails automatic responses in throttle and flight elevator signals, aligning the aircraft with the altitude setpoint. Consequently, these magnitudes in the digital twin closely mimic the behavior of these parameters present in real data. This is demonstrated in the enlarged section of Fig. 8a and b. These images were captured during a peak, possibly induced by a strong gust of wind, which caused a sudden increase in altitude.

As shown in Fig. 9, the aircraft reaches altitudes of 95 (m) and 135 (m), exceeding the respective reference altitude by approximately 15 meters in both experiments. At that moment, the throttle signal indicates a reduction in engine speed to nearly zero. This means the engine is not propelling the aircraft but rather gliding with the wind. Additionally, the elevator signal suggests a descent to reach the established altitude point. This behavior is evident in both the real and simulated aircraft, demonstrating that the digital twin closely replicates the behavior of the real aircraft. It's important to note that the digital twin shares identical characteristics and parameters with the real aircraft, particularly the same flight controller, the Pixhawk Cube, responsible for generating the mentioned control actions.

It's worth noting that the validated disturbance forces are estimated for steady-state flight, as illustrated in Fig. 8. The transition during takeoff and landing is not considered in the analysis. This is due to unexpected behaviors observed in both real and simulated experiments, likely caused by the same Ardupilot autopilot.

According to the meteorological information presented in [52], experiment 185, conducted in February, experienced moderately low wind currents. These moderate wind distur-

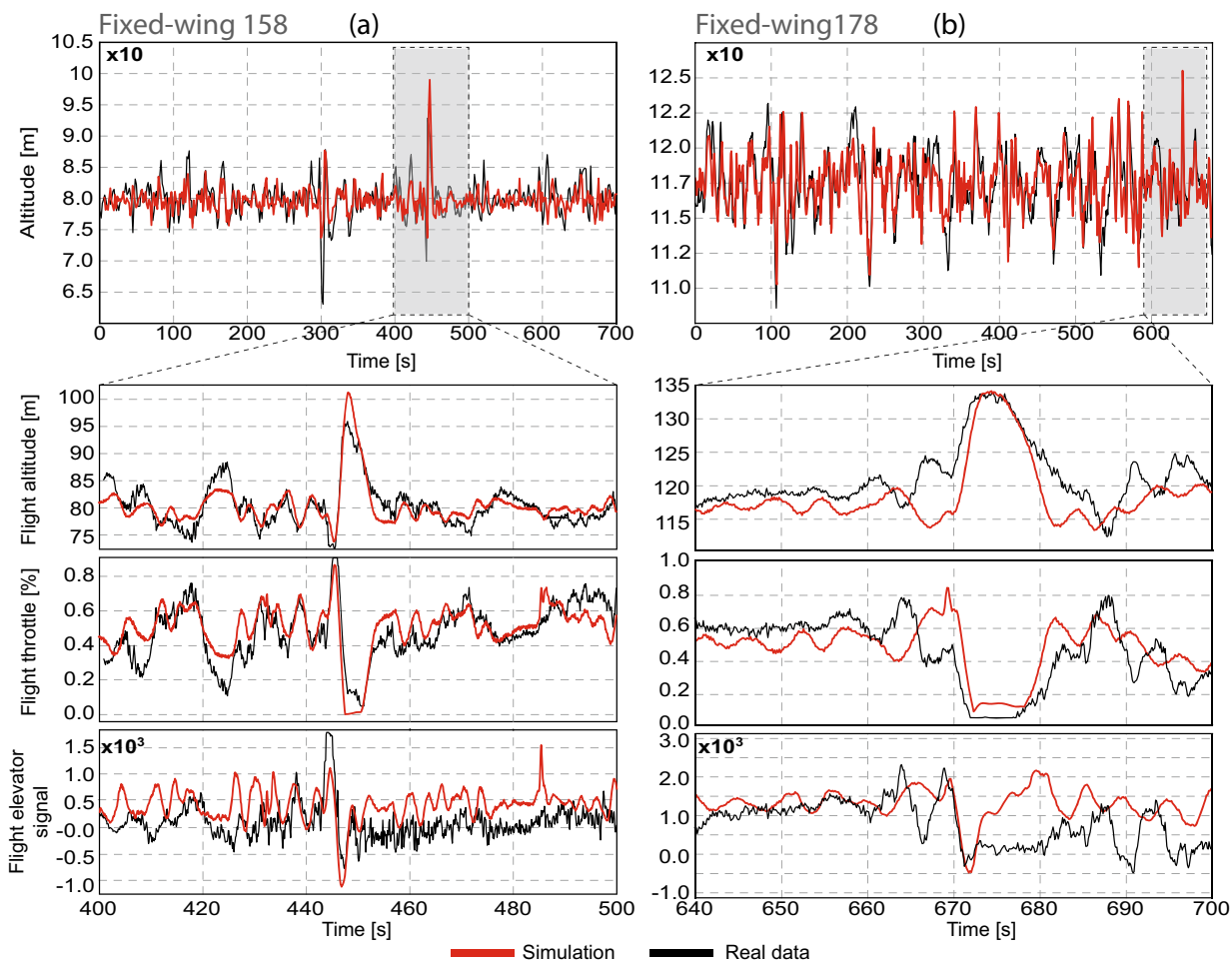

Fig. 9 Representative magnitude comparison between real flight and simulation with a fixed wing: (a) flight test 158 conducted in October, (b) flight test 178 conducted in December

bances resulted in average altitude deviations of 5 (m), with occasional peaks reaching up to 12 (m), as illustrated in Fig. 10a. Once again, altitude variations are accurately estimated along the steady-state stretch, especially during highly disturbed peaks highlighted in grey and represented in Fig. 10b. Similar to experiments 158 and 178, the simulated throttle and flight elevator signals closely respond to the real signals to position the aircraft at the 100 (m) altitude point. Most of the trajectory is influenced by moderate disturbances, which are closely emulated, as shown in the comparisons of real and simulated trajectories in Fig. 10c. It's important to note that the most significant differences between real and simulated flights occur during turns. This may be due to the impact of disturbance components in the X and Y axes, which were not accounted for in this study. However, as complementary data to this study, the mean square errors in the X and Y coordinates of flights 185,158 and 178 are presented in Table 5.

Flight conditions around Antisana are highly challenging due to variations in wind speed, which exert both favorable and adverse forces on the aircraft's movement. The results obtained provide insight into how the digital twin of the aircraft effectively demonstrates the aircraft's response to wind force. This information is invaluable when designing a fixed-wing aircraft, as they are typically designed for nearly ideal conditions, such as altitudes below 2000 meters above sea level or wind speeds below 8 (m/s). Designing an aircraft to operate at 4000 meters above sea level and speeds exceeding 8 (m/s) poses a significant challenge.

As shown earlier, it is possible to gain a clearer understanding of how an aircraft will perform in demanding environments such as Volcano Antisana in the High Andes. Recall that the proposed approach allows for the estimation of disturbances caused by changes in the wind. Therefore, a validated estimate can enable the analysis of different mission patterns for specific dates and times when real data was acquired. Estimating disturbances for different conditions will require new real flights.

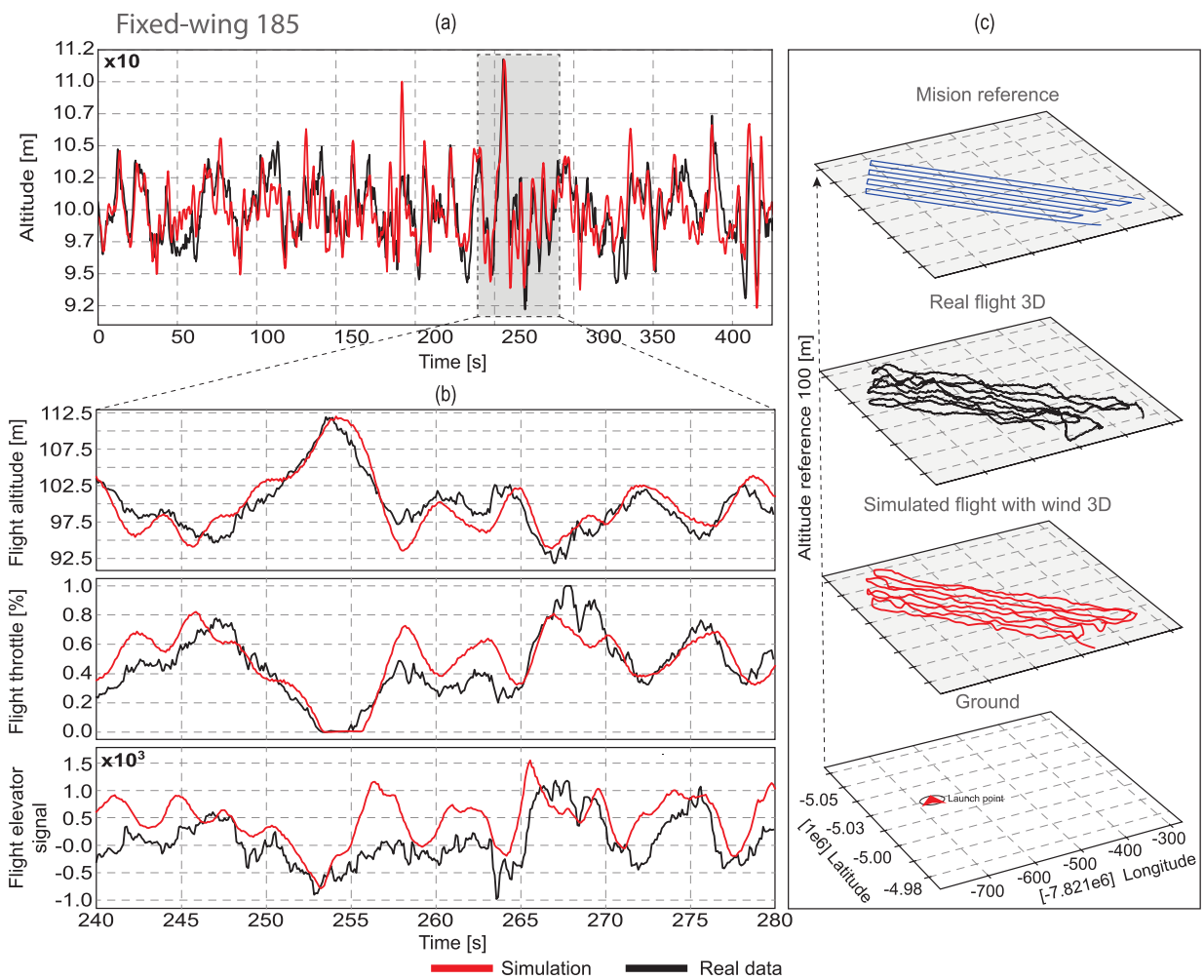

Fig. 10 Representative magnitude comparison between real flight and simulation for 185 mission conducted in February with a fixed wing: (a) altitude comparison, (b) related magnitude comparison, (c) trajectory comparison

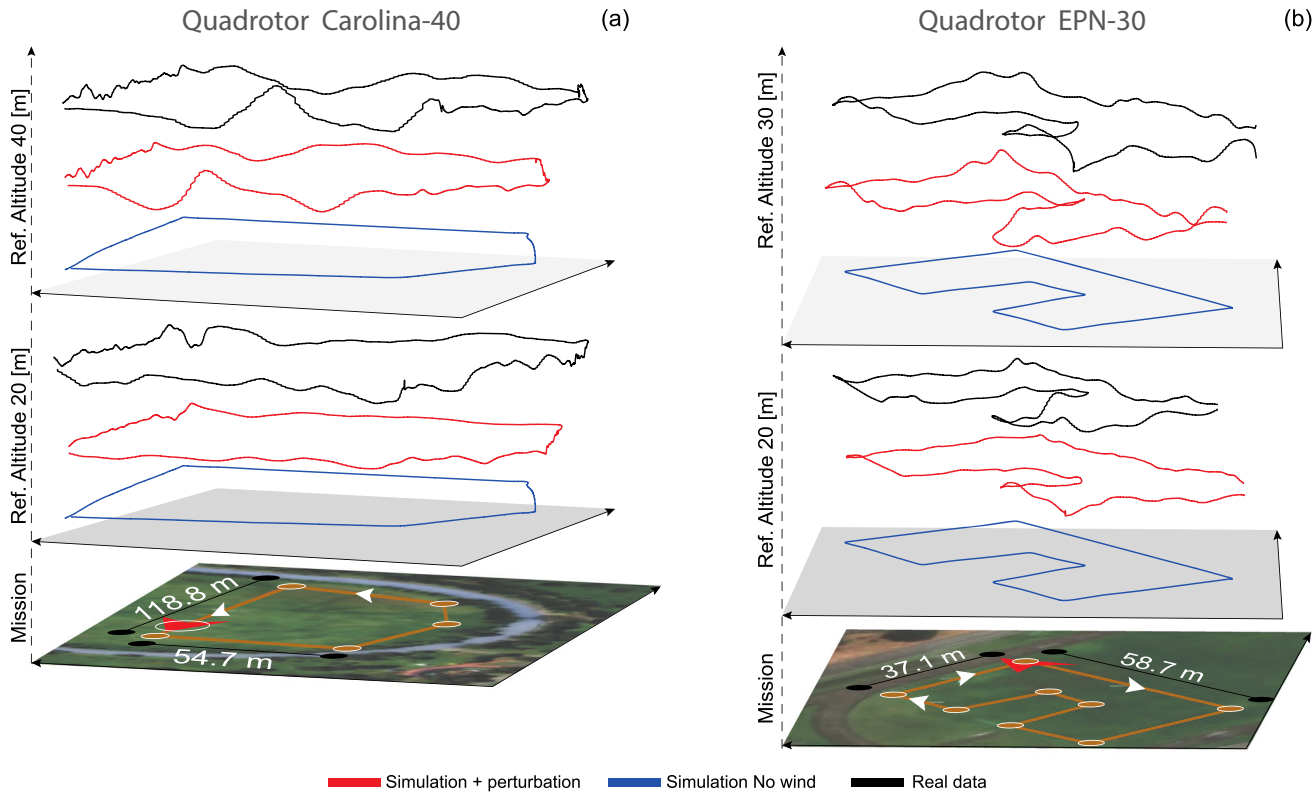

Fig. 11 Quadrotor experiments trajectories: (a) Carolina Park trajectories, (b) EPN trajectories

3.2 Quadrotor Experiments

The autonomy and configuration of a quadcopter pose challenges for its operation in monitoring activities in Antisana. The water present in clouds at altitudes above 4000 meters above sea level (masl) could potentially damage the exposed electronic components of the quadcopter, as illustrated in Fig. 7. The experimental validation involving the quadcopter aims to evaluate the wind disturbance estimation using a different digital twin model. All experiments were carried out in open urban spaces with moderate to strong winds, with speeds ranging between 5 and 7 m/s. The tests were performed on days without precipitation, minimizing potential environmental interference. It is important to note that most of the aerodynamic constants established for the digital twin match those of the "IRS-4" model, as well as the default configuration of the Ardupilot environment. The focus was on precisely tuning the PID controllers and refining the geo-

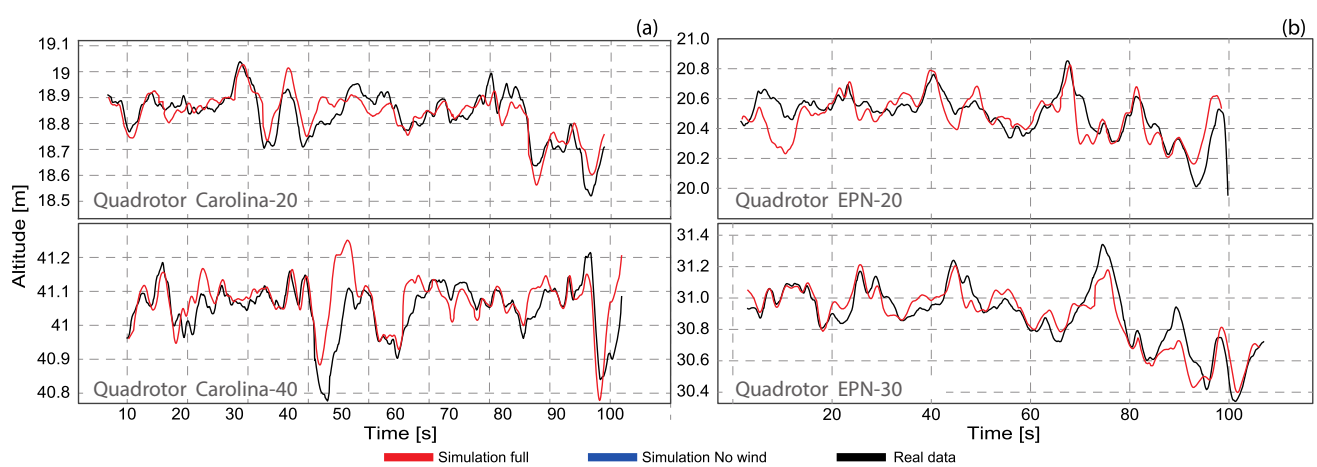

Fig. 12 Quadrotor altitude comparison:(a) Carolina Park experiment, (b) EPN experiment

Table 5 Error statistics for the fixed-wing Antisana missions and the quadrotor urban missions

Statistics	Fixed-wing			Quadrotor			
	Mission 185	Mission 178	Mission 158	Carolina-40	Carolina-20	EPN-30	EPN-20
RMSE in Z [m]	1.885	1.994	2.001	0.07	0.054	0.07	0.10
RMSE in X [m]	0.865	1.729	0.661	0.043	0.037	0.062	0.071
RMSE in Y [m]	0.373	1.248	0.247	0.039	0.027	0.055	0.036
Median [m]	-0.001	0.019	0.141	-0.013	-0.013	0.018	-0.015
75th percentile	1.165	1.163	1.243	0.056	0.046	0.078	0.068
25th percentile	-1.208	-1.222	-0.934	-0.01	-0.035	-0.027	-0.054

metric parameters to match the dimensions of our assembled model.

The experiments, labeled Carolina and EPN, were carried out on Parque Carolina, in the northern part of Quito, and on the soccer field of the EPN campus. These locations were selected due to their variable altitudes (2800 masl), allowing for an initial assessment of the impact of wind at high altitudes. In these experiments, the wind effects were significantly attenuated by the surrounding buildings, which provided partial shielding against the wind. As shown in Fig. 11, the experiments carried out in the Carolina park at 40 and on the EPN campus at 30 meters are significantly affected by wind gusts, and the simulated trajectories closely resemble the real ones. Altitudes below 20 meters are minimally affected by the wind, and most altitude deviations are caused by the quadcopter's movement. Simulated trajectories without wind show perfectly flat traces, despite sharp 90-degree turns, as observed in the blue traces in Fig. 11a and b.

Similarly to the previous section, the altitude variations observed in real quadcopter flights allow for the estimation of wind disturbance force along a single axis. This estimation aims to generate simulated altitude graphs that closely resemble the real ones, as illustrated in Fig. 12a and b. In this set of experiments, the altitude variation is less than 1 meter, and it is important to note that the most significant differences arise during turns. In these instances, the disturbance force may have larger X and Y components, which are eventually compensated for by the flight controller. This can be visualized in Table 5 where the RMSE values for the X and Y chords are shared as complementary data to this study. For a more comprehensive understanding of the virtual environment, readers are encouraged to watch the supplementary videos available in the 'video' folder of the shared link in Section 2.0. In this series of experiments, the terrain altitude is almost imperceptible; however, the presence of significant tall buildings is visualized as high blocks in the 3D scenario. These structures must be considered to avoid collisions during the mission.

Although tests conducted with fixed-wing aircraft demonstrate that the framework is adaptable to scenarios with more demanding conditions and higher-altitude flights, in the case of the quadcopter, it is important to consider its physical limitations, as these reduce its operational robustness. Unlike fixed-wing UAVs, quadcopters are more susceptible to sudden wind gusts, which can compromise flight stability. In addition, factors such as variations in air density at higher altitudes can affect its performance, particularly in terms of lift and battery efficiency. These aspects must be carefully evaluated before conducting long-duration or high-altitude missions.

Another aspect to consider is that, as mentioned in the previous section, the current disturbance estimation model only accounts for altitude variations (Z-axis) without consider-

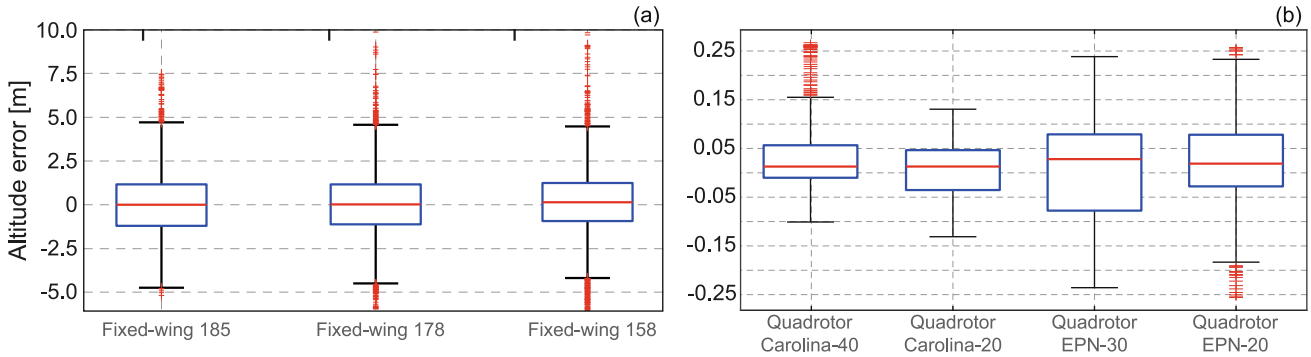

Fig. 13 Error boxplots for all flight tests: (a) Antisana missions with the fixed-wing plane, (b) urban quadrotor missions

ing wind effects on the X and Y axes. This simplification generates discrepancies between the simulated and real trajectories, especially during sharp turns. To address this limitation, an improved model that incorporates the three-dimensional estimation of the wind force is necessary.

3.3 Error Analysis

The validation error was addressed in relation to the change in altitude, a variable directly affected by our disturbance estimation approach as explained in the previous section. Variables such as flight acceleration or lift signal in the fixed-wing are automatically generated by the Ardupilot framework. The mean errors and root mean square errors calculated for the fixed-wing and quadcopter missions are shown in Fig. 13 and detailed in Table 5. It is evident that, in all experiments, the error deviations beyond the 75th and 25th percentiles are minimal. Consequently, the mean errors are hardly influenced by outliers in the estimation, and notably, there are almost no shifts in the estimates.

All flight tests conducted under strong winds and challenging weather conditions at Antisana show root mean square errors of approximately 1.8 (m), 1.9 (m), and 2.0 (m). These values denote errors of less than 8%, considering the largest altitude deviations and the total trajectory length.

In the quadcopter experiments, the errors are significantly lower, resulting in root mean square errors averaging approximately 0.07 (m). This substantial difference is attributed to the minimal impact of wind in urban areas, resulting in altitude deviations of approximately 1 (m). Overall, it is observed that the altitude error with the quadcopter averages around 9%, similar to the fixed-wing results.

Regardless of the UAV model, the external wind disturbance can be generalized to achieve accurate simulations that resemble the meteorological conditions at the date and time of the real flights. This capability serves as a valuable resource for analyzing UAV modifications and implementing new control strategies before their deployment in real conditions. In this way, the risk of damaging expensive onboard equipment is mitigated, and the integrity of the UAV is preserved. The estimation of the disturbance force is a work in progress, where a three-dimensional disturbance is considered the most suitable option to improve the accuracy of the current method.

3.4 Discussion

A deterministic simulation, despite accurately emulating flight controllers and terrain characteristics, produces stable and flat aircraft motion. However, estimating or generating a wind perturbation model remains complex due to the stochastic, multidirectional nature of wind. Wind gusts are irregular, short-lived, and prone to rapid directional changes, making them difficult to represent with deterministic force models in simulations. The proposed approach instead relies on discrete computations of external forces derived from real-world recorded data. This method simplifies the perturbation to its most representative force component, accounting for altitude variations caused by wind gusts. Future efforts will focus on estimating a three-dimensional force model to reduce positional errors and emulate lateral perturbations, which are relatively minor compared to altitude-related disturbances. Notably, lateral displacements are actively corrected by the flight controller to maintain the aircraft's intended mission trajectory.

Experimental validation in the Antisana region revealed a strong correlation between the digital twin's simulated trajectories and real flight data from fixed-wing UAV missions. By incorporating a wind perturbation model derived from real-world altitude variations, the dynamic response of the aircraft was replicated with remarkable precision for specific dates and weather conditions. During steady-state flight, the estimated perturbation force produced altitude profiles and control signals (elevator and throttle) closely matching those recorded in actual missions. Across all fixed-wing experiments, the root mean square error (RMSE) for altitude remained below 2 meters, corresponding to a relative error under 8%, even in severe environmental conditions with wind gusts exceeding 9 m/s. In contrast, quadrotor experiments exhibited significantly lower errors compared to fixed-wing tests, likely due to the quadrotor's inherent stability and the moderate, consistent wind currents typical of urban environments.

3.5 Quadrotor Challenges

Concerning the challenges encountered when using quadrotors, one of the most significant is related to battery limitations and inherent constraints in flight autonomy. Due to the quadrotor's lightweight structure, its control system frequently demands high-energy peaks when adjusting motor speeds for stability and maneuverability, particularly in the presence of strong wind gusts. This results in limited flight time, directly affecting the operational range and duration of remote sensing missions, especially in complex terrains such as the Andean mountains. Missions in these areas often require extended exploration periods and experience increased energy consumption caused by altitude variations and colder temperatures, which negatively impact battery performance. Potential solutions include optimizing flight paths, reducing payload weight, or utilizing more energy-efficient components; however, these options might compromise sensing capabilities or overall system robustness. Although our specific tests conducted in an urban environment successfully mitigated strong wind gusts due to controlled airflow patterns created by surrounding buildings,

the potential effects of stronger and less predictable gusts in open or high-altitude mountain areas remain a significant concern.

Furthermore, the sensitivity of PID controllers, especially regarding yaw stabilization, presented another considerable challenge. During our experiments, we observed minor oscillations and overshoots occurring when the UAV executed maneuvers involving curves, primarily due to inherent UAV dynamics and the continuous reorientation required for trajectory alignment. Interestingly, these behaviors were not noticeable during simulations, despite using identical PID tuning parameters, highlighting the significant influence of real-world conditions, such as subtle variations in UAV inertia and wind interactions, on controller performance. An additional limitation of our current approach is that it does not explicitly account for the effective aerodynamic area of the quadrotor. Consequently, tuning controller parameters to compensate for wind perturbations becomes sensitive and heavily reliant on empirical adjustments rather than a systematic analytical methodology. Future research will aim to develop a more comprehensive perturbation model that explicitly integrates the effective aerodynamic area of quadrotors or other UAV types. Such improvements will enable more precise predictions and adaptive control adjustments under variable wind conditions, ultimately enhancing the stability and reliability of UAV operations in complex environments.

3.6 Operative Cost Analysis

Our proposed approach significantly reduces operational costs by accurately emulating UAV behavior in the presence of wind gusts, which is particularly beneficial for remote sensing experiments conducted in distant mountain regions. Typically, testing UAVs in mountainous areas involves substantial expenses related to transportation, accommodation, and food, since these regions are often far from urban areas and challenging to access with regular vehicles. By replicating UAV behavior under complex wind conditions in a controlled simulation environment, our methodology eliminates the frequent need for physical travel to these remote test sites, thereby substantially reducing associated logistical expenses.

Additionally, our digital twin framework provides a safe environment for thoroughly evaluating new UAV designs and control strategies without risking physical damage to sensitive equipment. UAV experiments inherently carry a risk of crashes, particularly when testing novel designs, implementing new control actions, or encountering unpredictable wind conditions in complex terrains. In a simulated environment, crashes or critical operational failures do not result in hardware damage, significantly reducing financial burdens associated with repairing or replacing costly UAV components. Consequently, this approach not only facilitates rapid iterative design improvements but also ensures a more economical and efficient UAV development process.

4 Conclusions

The developed method demonstrated adequate accuracy based on information obtained from open sources. The results indicated that, for both fixed-wing aircraft and quadcopters, errors in the Z-axis were below 10%, enabling improved flight mission design in mountainous regions. In these areas, collisions are highly probable due to the terrain's topography and the uncertainties induced by onboard and external positioning systems. The method was validated in a rural environment near the Antisana volcano, where operational conditions were complex, and the effects of wind gusts had to be considered. In this context, the digital twin effectively incorporated external forces to realistically simulate wind effects, demonstrating its potential for real-world applications.

The digital twin, developed using open-source software, facilitated flight planning, mission execution, and pilot training, reducing operational costs and minimizing the risk of equipment failure. Additionally, avoiding the use of licensed software provides a competitive advantage for research institutions and NGOs, where costs must be kept to a minimum. Furthermore, the proposed approach for wind disturbance estimation proved to be generalizable to different UAV models. For quadcopters, similar error levels (9%) were observed, confirming the robustness of the method. This accuracy can be attributed to the integration of general aerodynamic principles, real flight data, and heuristic factors requiring mission-specific adjustments. However, the estimation of disturbance forces along a single axis was only effective for straight trajectories, while its accuracy decreased in curved paths. This limitation remains under investigation to develop a more precise estimation method.

The development of this project involved the use of Gazebo as a visualization environment, ArduPilot with custom firmware for fixed-wing aircraft, and ROS, enabling communication between different environments. To analyze the aircraft's aerodynamic behavior, results obtained with OpenFOAM, developed by [45], were used. The complete environment was created with the assistance of Blender. All these efforts successfully achieved the primary objective of generating a digital twin capable of visualizing the aircraft's behavior during specific missions with high accuracy-altitude errors relative to the ground were below 2 meters. This allows for the emulation of various missions while considering wind disturbances at the exact date and time when real-world data was acquired.

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Author Contributions Esteban Valencia contributed to the conception of the study, paper design, and fixed-wing monitoring missions. Francisco Toapanta, Gabriel Oña, Andrey Carrillo, Daniel Aláez, and Erick Loyaga developed the digital twin models. William Chamorro contributed to the experimental setup for the quadcopter case and the enhancement of figures. Patricio Cruz and Jackeline Abad supervised the creation of the digital twin model. Patrick Vandewalle contributed to the revision of the paper, networking, and securing funding for data gathering.

Declarations

Competing interests The authors have no relevant financial or non-financial interests to disclose.

Ethics approval This study did not require ethics approval.

Consent to participate Informed consent was obtained from all individual participants included in the study.

Consent for publication All participants in this study consented to publication.

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